

NW HAWAII BUOY 51001, HI, USA

WMO: 992020

Lat: 24.453N

Lon: 162.000W

Elev: 0

StdP: 101.33

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.3	19.9	N/A	N/A	N/A	N/A	N/A	N/A	14.3	21.2	13.7	21.4	8.1	30	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	1.4	28.2	N/A	27.9	N/A	27.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.2	11.4	10.7														
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	22.6	22.3	22.6	22.4	23.5	24.7	25.4	26.0	26.1	25.5	24.5	23.6
(7)		DBStd	1.69	1.12	1.20	1.37	0.99	1.13	0.66	0.66	0.80	0.81	0.95	0.80	0.98
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
(10)		CDD10.0	5146	391	344	389	372	417	442	476	494	482	481	435	421
(11)		CDD18.3	2104	133	111	131	122	159	192	218	236	232	222	185	163
(12)		CDH23.3	11381	168	101	198	81	476	1167	1657	2098	2124	1802	998	512
(13)		CDH26.7	328	0	0	1	0	0	0	8	106	128	83	2	1
(14)	Wind	WSAvg	6.9	6.9	6.8	7.0	7.3	6.1	6.9	7.5	7.0	6.3	6.8	7.0	7.1
(15)	Precipitation	PrecAvg	505	54	64	56	28	23	15	20	19	26	61	62	77
(16)		PrecMax	964	160	198	258	108	125	140	78	68	140	223	156	290
(17)		PrecMin	199	14	5	2	1	3	1	4	2	5	4	5	2
(18)		PrecStd	181	37	55	60	30	25	25	18	16	29	55	46	81
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	25.7	26.7	25.1	26.2	26.6	27.4	28.5	28.5	28.7	27.0	26.3
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(21)		2%	DB	24.9	25.1	26.0	24.6	25.7	26.3	26.9	28.0	28.0	28.2	26.5	25.8
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(23)		5%	DB	24.5	24.6	25.4	24.1	25.3	26.0	26.7	27.8	27.8	27.7	26.2	25.5
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(25)		10%	DB	24.2	24.0	24.1	23.9	25.1	25.8	26.4	27.4	27.5	27.0	25.8	25.1
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(28)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(30)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(32)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(34)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35)	Mean Daily Temperature Range	5% DB	MDBR	1.9	1.6	1.5	1.4	1.4	1.4	1.3	1.4	1.6	1.6	1.8	
(36)			MCDBR	2.4	1.6	1.6	1.5	1.5	1.4	1.5	1.3	1.5	1.8	2.1	2.5
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40)	Clear-Sky Solar Irradiance	taub	0.360	0.355	0.365	0.408	0.390	0.388	0.390	0.382	0.374	0.375	0.361	0.356	
(41)		taud	2.581	2.593	2.560	2.425	2.515	2.525	2.535	2.568	2.582	2.600	2.620	2.613	
(42)		Ebn at Noon	902	931	934	895	903	897	896	907	912	899	895	893	
(43)		Edh at Noon	89	94	101	118	108	106	105	101	98	92	85	84	
(44)	All-Sky Solar Radiation	RadAvg	4.02	4.66	5.49	6.47	7.00	7.36	7.17	6.92	6.29	5.07	4.14	3.73	
(45)		RadStd	0.19	0.41	0.49	0.37	0.41	0.18	0.23	0.23	0.26	0.26	0.21	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page